Script Format:
==========
Steps which will help the other users to understand our script easily.
1) Define shell (#!/bin/bash)
2) Comments (# comments)
3) Define vairbales (e.g name=sabair)
4) Commands (echo ,cp ,grep etc.)
5) Statements (if,while,for etc)

1.Script 1

```
[[ec2-user@ip-172-31-28-176 ~]$ mkdir -p /kamaal/scripts
mkdir: cannot create directory '/kamaal': Permission denied
[[ec2-user@ip-172-31-28-176 ~]$ sudo su -
[[root@ip-172-31-28-176 ~]# mkdir -p /kamaal/scripts
[root@ip-172-31-28-176 ~]# cd /kamaal/scripts
[[root@ip-172-31-28-176 scripts]# vi helloworld.bash
[[root@ip-172-31-28-176 scripts]# ./helloworld.bash
-bash: ./helloworld.bash: Permission denied
[[root@ip-172-31-28-176 scripts]# chmod 777 helloworld.bash
[[root@ip-172-31-28-176 scripts]# ./helloworld.bash
kamaal
[root@ip-172-31-28-176 scripts]#
```

2.created command.script

```
[[root@ip-172-31-28-176 scripts]# vi commands.bash
[[root@ip-172-31-28-176 scripts]# ./commands.bash
-bash: ./commands.bash: Permission denied
[[root@ip-172-31-28-176 scripts]# chmod 777 commands.bash
[[root@ip-172-31-28-176 scripts]# ./commands.bash
/kamaal/scripts
ls: cannot access 'whoami': No such file or directory
Thu Oct 15 09:31:06 UTC 2026
    October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

[root@ip-172-31-28-176 scripts]#
```

3.created a admin.bash script file

```
[root@ip-172-31-28-176 scripts]# whoami
root
[root@ip-172-31-28-176 scripts]# vi admin.bash
[root@ip-172-31-28-176 scripts]# chmod 755 admin.bash
[root@ip-172-31-28-176 scripts]# ./admin.bash

top - 09:41:09 up  3:21,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 116 total,   1 running, 115 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.2 st
MiB Mem :    913.1 total,    328.4 free,    192.4 used,    392.3 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    583.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  174464  18320  11264 S   0.0   2.0   0:02.71 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.01 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.15 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.24 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.04 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.07 migration/1
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.15 ksoftirqd/1
     24 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
```

add head-10 so its shows only 10 lines

```
#!/bin/bash
#
top | head -10
df -h
free -m
uptime
iostat
~
~
~
~
~
~
~
~
~
~
~
~
```

```
top - 09:46:08 up  3:26,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 117 total,   1 running, 116 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  3.1 sy,  0.0 ni, 96.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    913.1 total,    330.0 free,    190.6 used,    392.5 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    584.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  174464  18320  11264 S   0.0   2.0   0:02.74 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           457M     0  457M   0% /dev/shm
tmpfs           183M  452K  183M   1% /run
efivarfs        128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1  8.0G  1.7G  6.3G  21% /
tmpfs           457M     0  457M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs            92M     0   92M   0% /run/user/1000
               total        used        free      shared  buff/cache   available
Mem:             913         190         329           0         392         584
Swap:              0           0           0
 09:46:08 up  3:26,  2 users,  load average: 0.00, 0.00, 0.00
Linux 6.18.54-121.166.amzn2023.x86_64 (ip-172-31-28-176.ec2.internal)   10/15/26        _x86_64_        (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
          0.21    0.01    0.16    0.03    0.19   99.40

Device             tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1           1.90        19.38        70.48         0.00     240463     874576          0


[root@ip-172-31-28-176 scripts]#
```

4.Created a variables.bash script file

```
[root@ip-172-31-28-176 scripts]# vi variables.bash
[root@ip-172-31-28-176 scripts]# chmod 755 variables.bash
[root@ip-172-31-28-176 scripts]# ./variables.bash
Run Variables tasks

/kamaal/scripts
a  admin.bash  b  c  commands.bash  helloworld.bash  variables.bash
root
Thu Oct 15 09:53:35 UTC 2026
                                2023

       January               February                March
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7             1  2  3  4             1  2  3  4
 8  9 10 11 12 13 14    5  6  7  8  9 10 11    5  6  7  8  9 10 11
15 16 17 18 19 20 21   12 13 14 15 16 17 18   12 13 14 15 16 17 18
22 23 24 25 26 27 28   19 20 21 22 23 24 25   19 20 21 22 23 24 25
29 30 31               26 27 28               26 27 28 29 30 31

        April                  May                   June
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
                   1          1  2  3  4  5  6                1  2  3
 2  3  4  5  6  7  8    7  8  9 10 11 12 13    4  5  6  7  8  9 10
 9 10 11 12 13 14 15   14 15 16 17 18 19 20   11 12 13 14 15 16 17
16 17 18 19 20 21 22   21 22 23 24 25 26 27   18 19 20 21 22 23 24
23 24 25 26 27 28 29   28 29 30 31            25 26 27 28 29 30
30
        July                  August               September
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
                   1             1  2  3  4  5                   1  2
 2  3  4  5  6  7  8    6  7  8  9 10 11 12    3  4  5  6  7  8  9
 9 10 11 12 13 14 15   13 14 15 16 17 18 19   10 11 12 13 14 15 16
16 17 18 19 20 21 22   20 21 22 23 24 25 26   17 18 19 20 21 22 23
23 24 25 26 27 28 29   27 28 29 30 31         24 25 26 27 28 29 30
```

5.Created a input.bash script file

```
[root@ip-172-31-28-176 scripts]# vi input.bash
[root@ip-172-31-28-176 scripts]# chmod 755 input.bash
[root@ip-172-31-28-176 scripts]# chmod 755 input.bash^C
[root@ip-172-31-28-176 scripts]# ./input.bash
Hello, my name is sharaf kamaal

what is your name?
sharaf

Hello sharaf

[root@ip-172-31-28-176 scripts]#
```

6.Created a output.bash script file

```
a=`hostname`
echo Hello, my name is $a
echo
echo what is your name?
read b
echo
echo Hello $b
echo
~
~
~
~
```

```
[root@ip-172-31-28-176 scripts]# vi output.bash
[root@ip-172-31-28-176 scripts]# chmod 755  output.bash
[root@ip-172-31-28-176 scripts]# ./output.bash
Hello, my name is ip-172-31-28-176.ec2.internal

what is your name?
sharaf

Hello sharaf

[root@ip-172-31-28-176 scripts]#
```

7.Created a Conditions/ifelse.bash script file

```
[[root@ip-172-31-28-176 scripts]# vi output.bash
[[root@ip-172-31-28-176 scripts]# vi ifelse.bash
[[root@ip-172-31-28-176 scripts]# cat ifelse.bash
#!/bin/bash
count=100
if [ $count -eq 100 ]
then
echo count is 100
else
echo count is not 100
fi
[[root@ip-172-31-28-176 scripts]# chmod 755 ifelse.bash
[[root@ip-172-31-28-176 scripts]# ,/ifelse.bash
-bash: ,/ifelse.bash: No such file or directory
[[root@ip-172-31-28-176 scripts]# ./ifelse.bash
count is 100
[root@ip-172-31-28-176 scripts]#
```

```
[[root@ip-172-31-28-176 scripts]# vi ifelse.bash
[[root@ip-172-31-28-176 scripts]# cat ifelse.bash
#!/bin/bash
if [ -e /home/error.txt]
then
echo "File exist"
else
echo "File does not exist"
fi

[[root@ip-172-31-28-176 scripts]# chmod 755 ifelse.bash
[[root@ip-172-31-28-176 scripts]# ./ifelse.bash
./ifelse.bash: line 2: [: missing `]'
File does not exist
[root@ip-172-31-28-176 scripts]#
```

8. Created a multipleoption.bash script file

```
[root@ip-172-31-28-176 scripts]# vi multipleoption.bash
[root@ip-172-31-28-176 scripts]# vi multipleoption.bash
[root@ip-172-31-28-176 scripts]# ./multipleoption.bash

Please choose one of the options below

a = Display Date and Time
b = List files and directories
c = List users logged in
d = Check System uptime

a
Thu Oct 15 10:36:38 UTC 2026
[root@ip-172-31-28-176 scripts]# ./multipleoption.bash

Please choose one of the options below

a = Display Date and Time
b = List files and directories
```

9.Created a do-while.bash script file

```
 10:36:45 up  4:17,  2 users,  load average: 0.00, 0.00, 0.00
[root@ip-172-31-28-176 scripts]# vi do-while.bash
[root@ip-172-31-28-176 scripts]# cat do-while.bash
#!/bin/bash
count=0
num=10
while [ $count -lt 10 ]
do
        echo
        echo $num seconds left to stop this process $1
        echo
        sleep 1
num=`expr $num - 1`
count=`expr $count + 1`
done
echo
echo $1 process is stopped!!!
echo

[root@ip-172-31-28-176 scripts]# chmod 755 do-while.bash
[root@ip-172-31-28-176 scripts]# ./do-while.bash

10 seconds left to stop this process


9 seconds left to stop this process


8 seconds left to stop this process


7 seconds left to stop this process
```

10.Created a exit.bash script file

```
[[root@ip-172-31-28-176 scripts]# vi exit.bash
[[root@ip-172-31-28-176 scripts]# ./exit.bash
hello
0
[root@ip-172-31-28-176 scripts]#
```

```
[[root@ip-172-31-28-176 scripts]# vi exit.bash
[[root@ip-172-31-28-176 scripts]# ./exit.bash
ls: cannot access '/home/kamaal/check': No such file or directory
File does not exist
[root@ip-172-31-28-176 scripts]#
```